

# Motorola XTL 2500 → PC Direct Audio

Fixed receive station (indoor, not a vehicle). Parts, pictures, what plugs into what, and step-by-step instructions. Radio model M21URM9PW1AN (700/800 MHz, 35 W). October 2026.

**What you are building.** A listening station. The radio sits indoors on a desk or shelf, receives the P25 traffic and decodes it to normal speech. That speech leaves the radio through a **wire** on its rear accessory socket (not through a microphone), goes into a USB sound card, and the PC (or Raspberry Pi) turns it into text and fires your triggers. The radio never needs to transmit for this.

**There are two completely separate systems.**

**1. POWER** — wall outlet → 13.8 V power supply → Motorola power cable → radio. *The radio cannot be powered from USB.* At a station the supply runs all the time.

**2. AUDIO** — radio rear socket → HLN6863B plug → capacitor → 3.5 mm cable → ground-loop isolator → USB sound card → PC.

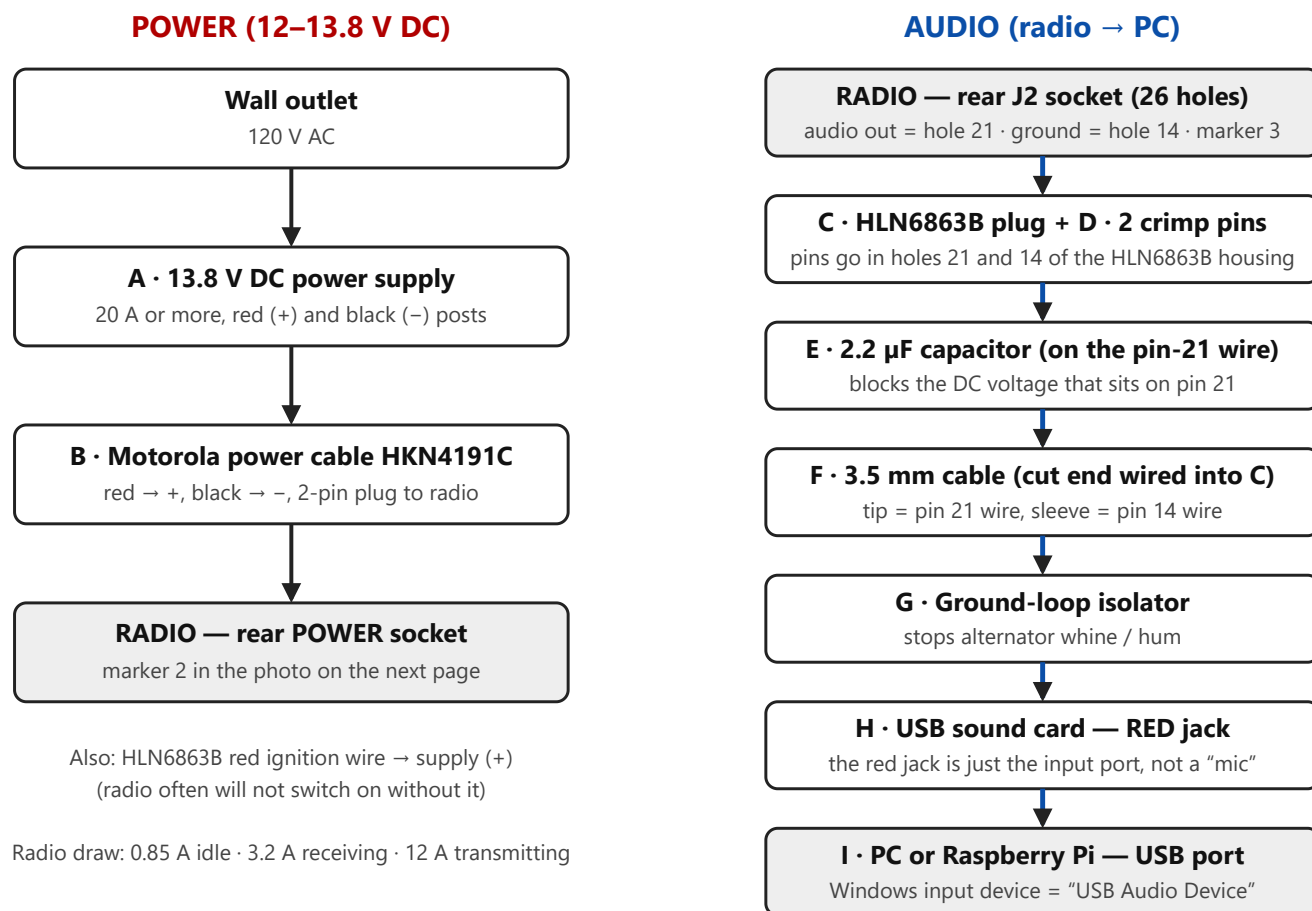


Figure 1. Everything in one picture. Letters A–I match the parts list. Blue arrows are the audio path.

# 1. Your radio — where things plug in

This is the back of your XTL 2500. Only two sockets matter for this build: the **POWER** socket (2) and the **J2 accessory** socket (3). The antenna (1) must also be connected for the radio to receive.



| # | Socket                  | What plugs in                                                                                                                   |
|---|-------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1 | Antenna (mini-UHF)      | Part <b>J</b> , a 700/800 MHz antenna. Indoors: a mag-mount on a steel plate near a window. The radio hears nothing without it. |
| 2 | POWER (2-pin)           | Part <b>B</b> , the Motorola power cable HKN4191C. Comes from the 13.8 V supply (part <b>A</b> ).                               |
| 3 | J2 accessory (26 holes) | Part <b>C</b> , the HLN6863B plug. Audio for the PC comes out of <b>hole 21</b> ; <b>hole 14</b> is its ground.                 |

Figure 2. Rear of your radio. “J2” is simply Motorola’s name for socket 3. “Pin 21 / pin 14” are two of the 26 holes inside it.



Figure 3. Front. The round jack at bottom-left is for the hand microphone — **not used** for the PC. The power button (⏻) is the third small round button left of the display.

## Never do these

- Never put a wire in holes **20** or **26** (speaker) and never let them touch ground. That destroys the radio’s audio amplifier.
- Never touch the J2 socket or plug while the radio has power.
- Never power the radio from a USB port, phone charger or small wall adapter. It needs 12–13.8 V and up to 12 A.
- Never press the microphone PTT (transmit) without an antenna connected.

**Why hole 21?** It is "RX filtered audio" — a fixed-level copy of whatever the radio receives. It ignores the volume knob, so the station needs no speaker at all; the PC still hears everything.

## 2. Parts to buy (with pictures)

Prices are October 2026, US. "Search" is the exact text to type into Amazon / eBay / Google.

### Power



#### **A** 13.8 V DC power supply, 20 A or more

Search: 13.8V 25A power supply ham radio — e.g. Samlex SEC-1223 (23 A), Astron SS-30M, Pyramid PS26KX

Price: \$70–160

Connects to: wall outlet (AC in) → part B on its red/black posts

Why: receiving takes 3.2 A, but a radio on a trunked system sends short automatic registration bursts at 12 A. 20 A keeps the voltage steady during those.



#### **B** Motorola power cable HKN4191C

Search: HKN4191C (10 ft, 12 AWG, 20 A fuse). 20 ft version: HKN4192C .

Price: ~\$28

Connects to: 2-pin plug → radio POWER socket (2). Bare ends → red (+) / black (–) posts on part A.

Why: it is the only plug that fits the radio's power socket, and it carries the fuse.

### Audio — radio side



#### **C** Motorola HLN6863B rear accessory connector

Search: HLN6863B (also HLN6863 / HLN6863A). "Mid-power rear ignition cable, XTL APX".

Price: \$39–44 new (North Georgia Comm., Atlantic Radio, Magnum), ~\$12 used on eBay

Connects to: the black block plugs onto radio socket J2 (3). Its red fused wire → supply (+) (ignition sense).

Why: it is the plug that fits J2. You add two pins (part D) for the PC audio.



#### **D** Crimp pins for the HLN6863B housing (need 2, buy 4)

Search: 3980034F01 or 3980034F04 (Wiscomm ~\$1.40 each) — or eBay HLN6961 XTL APX accessory connector pins

Price: ~\$6

Connects to: crimped onto the two wires of part F, then pushed into holes 21 and 14 of part C.

Why: ordinary D-sub pins do not fit this housing.



**E 2.2 µF non-polarized (bipolar) capacitor, 50 V**

Search: 2.2uF bipolar capacitor 50V (bag of 10–20). A 2.2uF film capacitor also works.

Price: ~\$6

Connects to: spliced *into* the pin-21 (signal) wire between part C and the 3.5 mm plug. Not on the ground wire.

Why: pin 21 carries a DC voltage with the audio; the capacitor lets audio through and blocks the DC.



**F Shielded 3.5 mm TRS cable, 3–6 ft (you cut one end off)**

Search: 3.5mm TRS male to male shielded audio cable 6ft

Price: ~\$7

Connects to: cut end → pins (D) → HLN6863B holes 21/14. Plug end → isolator (G).

Why: gives you a ready-made 3.5 mm plug and shielded wire in one buy.

## Audio — PC side



**G 3.5 mm ground-loop isolator**

Search: Cable Matters 3.5mm ground loop isolator (Amazon B08G9J74VX)

Price: ~\$12

Connects to: 3.5 mm plug of part F goes in one end; other end goes into the RED jack of part H.

Why: radio and PC are on different grounds; without it you get a loud hum / alternator whine.



**H USB sound card — UGREEN 30724**

Search: UGREEN USB audio adapter 30724 (Amazon B01N905VOY). Alternative with true line-in: Behringer UCA202 (~\$30).

Price: ~\$10

Connects to: isolator (G) → **RED** jack. USB end → PC / Pi.

Why: turns the audio wire into a USB input device. No drivers on Windows or Pi.

## Antenna



**J 700/800 MHz antenna with mini-UHF plug**

Search: 800 MHz mag mount antenna mini-UHF (Laird, Tram, Browning, ~\$20–35). Put the magnet on a steel plate or cookie sheet (ground plane) near a window.

Better reception: outdoor 800 MHz base antenna + mini-UHF male to N female adapter.

Connects to: radio antenna socket (1).

Why: no antenna = nothing to transcribe. Higher and closer to a window = more traffic heard.

### Optional: speaker for the room

The XTL 2500 has no built-in speaker. The PC does not need one. If you also want to hear traffic in the room, a Motorola external speaker (e.g. HSN4031) goes on the speaker leads already in the HLN6863B — not on holes 21/14.

## Tools and small stuff

| Item                                                    | Notes                                                                                                              |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Wire stripper / cutter                                  | For the thin 3.5 mm cable (22–26 AWG).                                                                             |
| Small needle-nose pliers or crimp tool                  | To close the crimp pins (D). Pliers work if you squeeze carefully.                                                 |
| Soldering iron + solder (or 2 small butt-splice crimps) | To join the capacitor into the wire. Twist-and-tape is fine for a first test; solder it for the permanent station. |
| Heat-shrink tubing or electrical tape                   | Cover the capacitor joints so they cannot touch anything.                                                          |
| Multimeter (~\$15, strongly recommended)                | To confirm which wire is tip and which is sleeve, and to check the supply voltage.                                 |
| Steel plate / cookie sheet                              | Ground plane for the mag-mount antenna. About 12 × 12 in.                                                          |

**Cost:** audio parts ≈ \$80 (≈ \$50 with a used HLN6863B). Power supply + cable ≈ \$100–190. Antenna ≈ \$25–35.

### 3. Connection map — what touches what

| #  | This...                                       | ...plugs into / joins...                     | Details                                                                                      |
|----|-----------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------|
| 1  | Power supply <b>A</b> — AC cord               | Wall outlet                                  | Keep the supply switched OFF until step 11.                                                  |
| 2  | Power cable <b>B</b> — bare RED end           | Supply <b>A</b> — red (+) post               | Fuse holder stays on the red wire.                                                           |
| 3  | Power cable <b>B</b> — bare BLACK end         | Supply <b>A</b> — black (–) post             |                                                                                              |
| 4  | Power cable <b>B</b> — 2-pin plug             | Radio POWER socket (2)                       | Fits one way only; push until it latches.                                                    |
| 5  | HLN6863B <b>C</b> — black block               | Radio J2 socket (3)                          | Only after the pins (D) are installed. Tighten its screw.                                    |
| 6  | HLN6863B <b>C</b> — red fused “ignition” wire | Supply <b>A</b> — red (+) post               | Tells the radio “the vehicle is on”. At a station it just stays on the (+) post permanently. |
| 7  | 3.5 mm cable <b>F</b> — inner (signal) wire   | Capacitor <b>E</b> — lead 1                  | Cut the signal wire; capacitor bridges the cut.                                              |
| 8  | Capacitor <b>E</b> — lead 2                   | Crimp pin <b>D</b> → HLN6863B hole <b>21</b> | Either lead can face the radio (bipolar).                                                    |
| 9  | 3.5 mm cable <b>F</b> — shield (ground) wire  | Crimp pin <b>D</b> → HLN6863B hole <b>14</b> | No capacitor on this wire.                                                                   |
| 10 | 3.5 mm cable <b>F</b> — plug end              | Isolator <b>G</b> — female end               |                                                                                              |
| 11 | Isolator <b>G</b> — male plug                 | USB sound card <b>H</b> — RED jack           | Red = input. Green = headphones (not used).                                                  |
| 12 | USB sound card <b>H</b> — USB-A               | PC / Raspberry Pi <b>I</b>                   | Shows up as “USB Audio Device”.                                                              |
| 13 | Antenna <b>J</b> — mini-UHF plug              | Radio antenna socket (1)                     | Mag-mount on a steel plate by a window; or outdoor antenna via adapter.                      |

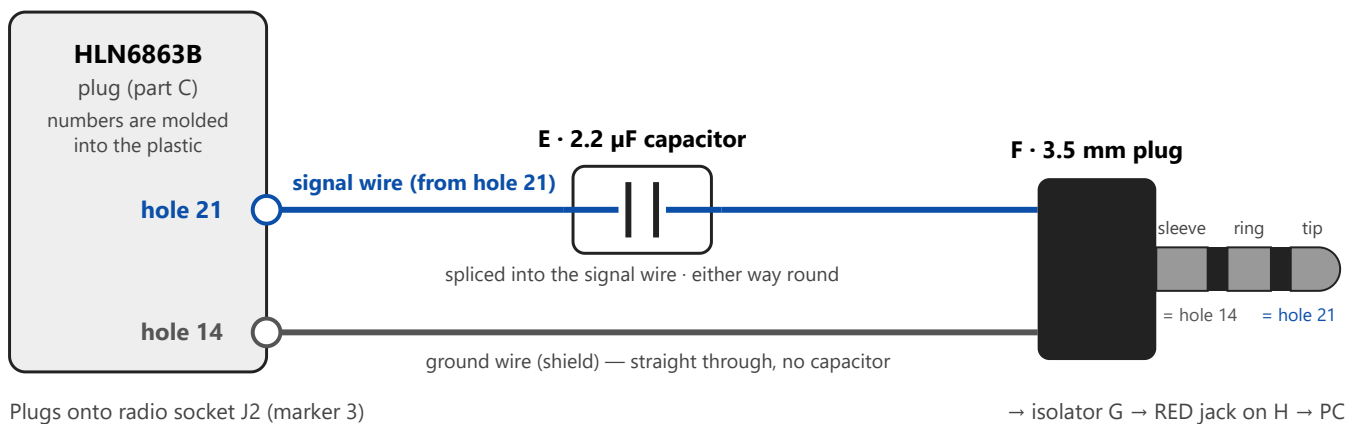


Figure 4. The one cable you build. Blue = signal from hole 21, through the capacitor, to the plug’s tip. Grey = ground from hole 14 straight to the plug’s sleeve. If your cable has two inner wires (stereo), twist them together and treat them as one signal wire.

## 4. Step-by-step build

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### Part A — Build the audio cable (nothing plugged in yet)

#### Step 1 — Cut the 3.5 mm cable

- Measure about **18 inches** from one plug and cut the cable there. You will use that 18-inch piece (one plug + wire). The rest is spare.
- Strip **1 inch** of the outer black jacket from the cut end. Be gentle — the wires inside are thin.
- Inside you will find a **bare braided or foil shield** wrapped around **one or two insulated wires** (often red and white).
- Twist the shield strands together into one wire. If there are two insulated wires, strip ¼ in from each and twist them together — they become one **signal** wire.

#### Step 2 — Confirm which wire is which (multimeter, continuity / beep mode)

- One probe on the plug's **tip** (the very end), other probe on the **signal** wire → it should beep.
- One probe on the plug's **sleeve** (the long metal part next to the cable), other probe on the **shield** wire → beep.
- Signal to shield → **no** beep. If it beeps, strands are touching; separate them.
- No multimeter? The shield is always the bare braid; the insulated wires are always signal.

#### Step 3 — Splice the 2.2 µF capacitor into the signal wire

- About **3 inches** from the cut end, cut the **signal wire only**. Strip ¼ in from both new ends. Leave the shield wire whole.
- Slide a piece of heat-shrink onto each end before joining (easy to forget).
- Twist one capacitor lead onto one cut end, the other lead onto the other cut end. Solder both joints (or use two small butt-splice crimps). Trim the leads short.
- Direction does **not** matter — a bipolar capacitor has no + or −.
- Shrink or tape each joint so no bare metal is exposed. The capacitor is now “in series on pin 21”, exactly as the notes say.

#### Step 4 — Crimp the two Motorola pins

- Strip ⅛–3/16 in of insulation from the end of the signal wire (the end past the capacitor) and from the shield wire.
- Lay the bare copper in the small barrel of a pin; fold the small tabs over the copper, then the larger tabs over the insulation. Squeeze firmly with the crimp tool or pliers.
- Pull gently on each wire. If a pin slides off, re-crimp. A dab of solder on the copper crimp adds strength.

#### Step 5 — Push the pins into the HLN6863B housing

- Look at the **wire side** (back) of the black HLN6863B block. Tiny numbers are molded next to the holes. Find **21** and **14**. Some positions already have wires (ignition, speaker) — leave them alone.
- Push the **signal** pin into hole **21** until it clicks. Push the **shield** pin into hole **14** until it clicks.
- Tug each wire lightly — it must not come back out.
- Check with the meter: signal pin ↔ plug tip = **no beep** (the capacitor blocks DC — that is correct); shield pin ↔ plug sleeve = beep.

## Part B — Plug everything into the radio (power still OFF)

### Step 6 — Connect the HLN6863B to the radio

- Push the HLN6863B block onto the **J2 socket** (marker 3, far right on the back of the radio). It only goes on one way. Tighten its screw finger-tight.
- Keep the 3.5 mm plug end free for now. Do not plug it into anything that is powered.

### Step 7 — Connect the power cable to the radio

- Push the 2-pin plug of the HKN4191C into the **POWER socket** (marker 2) until it latches.
- Check the fuse holder on the red wire has its 20 A fuse in it.

### Step 8 — Connect the antenna

- Put the mag-mount on the steel plate, as high and as close to a window as the cable allows (away from the PC and the power supply).
- Screw its mini-UHF plug onto the antenna socket (marker 1). Finger-tight is enough. Without an antenna you will hear almost nothing.

### Step 9 — Build the chain to the PC

- 3.5 mm plug (from Part A) → **female** end of the ground-loop isolator.
- Isolator **male** plug → **RED** jack of the UGREEN adapter. (Green is for headphones — ignore it.)
- UGREEN USB end → any USB port on the PC / Pi.

## Part C — Power up

### Step 10 — Wire the power supply (supply switched OFF)

- HKN4191C **red** bare end → supply **red (+)** post. **Black** bare end → supply **black (-)** post.
- HLN6863B **red fused ignition wire** → also onto the supply **red (+)** post (two wires on the same post is fine). This is the “ignition sense” the radio looks for.
- Plug the supply into the wall.

### Step 11 — Switch on

- Switch the supply on. The meter (or a multimeter on the posts) should read about **13.8 V**.
- Press the radio’s **power button** (⏻, third small button left of the display). Display lights, radio beeps.
- Select the channel / talkgroup you want to monitor. The volume knob does not matter for the PC — pin 21 is fixed level.

## Part D — PC setup

### Step 12 — Windows (station PC)

- **Settings** → **System** → **Sound** → **Input**. Choose **USB Audio Device** (sometimes “USB PnP Sound Device”).
- Set input volume to about **30 %** to start. If there is a “Microphone boost” option under device properties, set it to **0 dB**.
- Watch the input level meter during a transmission. It should bounce clearly but **not** hit the top. Raise or lower the input volume until it peaks around  $\frac{2}{3}$ .

### Step 13 — Point the listener at the USB device

- In `device/config.json` set `input_device` to the USB Audio Device (name or index — whatever your listener accepts).
- Run `python listener.py`. Check its level readout during a transmission; adjust the Windows input volume if it clips.
- Raspberry Pi: `arecord -l` lists the card number of the USB device; put that in `config.json` the same way.
- For a station that runs unattended, set the listener to start automatically at login (Task Scheduler on Windows, a systemd service on the Pi).

#### Station notes

- **Placement:** radio on a hard surface with air around the heat-sink fins; antenna by a window; supply and PC on the same power strip (less hum).
- **Always on:** supply stays on, ignition wire stays on (+), radio stays on. It uses about 45 W while listening.
- **Keep the isolator:** the radio supply and the PC are still two different ground paths through the mains.
- **Trunked systems:** if the radio is programmed for a trunked P25 system, it transmits short registration bursts on its own. That is normal for a radio authorized on that system — and the reason for the 20 A supply and the “antenna always connected” rule.
- **Transcription and triggers** happen entirely on the PC. Nothing in the hardware changes when you add or change keywords.

### Troubleshooting

| Problem                                           | Check                                                                                                                                                                                                                                                       |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Radio is on but never receives anything           | Antenna plugged in and on its steel plate? Try another spot near a window. Right channel / talkgroup selected? Signal bars on the display?                                                                                                                  |
| Radio does nothing when I press power             | Ignition wire from the HLN6863B must be on the supply (+). Supply switched on and reading 13–14 V? Fuse in the HKN4191C blown? Power plug latched?                                                                                                          |
| Windows meter does not move during a transmission | Pins fully seated in holes 21 and 14? Capacitor joints solid (meter: pin 21 ↔ capacitor near lead = beep; capacitor far lead ↔ plug tip = beep)? Plug pushed all the way into the isolator and the isolator into the RED jack? Right input device selected? |
| Still silent, everything checks out               | In CPS: Radio Configuration → Radio Wide → Advanced, check the rear accessory audio options. On most XTLs pin 21 is always live.                                                                                                                            |
| Loud hum or whine                                 | Isolator in the chain? Try the PC on battery for a moment — if the hum disappears, it is a ground loop; keep the isolator and keep the supply and PC on the same outlet strip.                                                                              |
| Too quiet                                         | Raise Windows input volume. Still low → try the CPS Accessory → Rear RX audio gain.                                                                                                                                                                         |
| Distorted / clipping                              | Lower Windows input volume; make sure mic boost is 0 dB.                                                                                                                                                                                                    |
| Constant hiss between transmissions               | Normal when the squelch opens. The software audio gate (next to-do in the notes) handles this.                                                                                                                                                              |

### Before you buy — checklist

- ☐ A — 13.8 V supply, 20 A+

- ☐ B — HKN4191C power cable
- ☐ C — HLN6863B rear accessory connector
- ☐ D — 4 × crimp pins 3980034F01 / F04
- ☐ E — 2.2  $\mu$ F bipolar capacitor, 50 V
- ☐ F — 3.5 mm TRS male-to-male shielded cable
- ☐ G — Cable Matters 3.5 mm ground-loop isolator
- ☐ H — UGREEN 30724 USB audio adapter
- ☐ J — 800 MHz mag-mount antenna with mini-UHF plug + steel plate
- ☐ Tools: stripper, pliers/crimp tool, soldering iron or butt splices, heat-shrink, multimeter